

Product Requirements Document

Artist Digital Commerce & Engagement Platform

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Executive Summary

This platform provides artists with a unified ecosystem to manage and monetize their creative output through live events, music distribution, and merchandise sales. By consolidating these revenue streams into a single platform, artists gain ownership of fan relationships, data insights, and direct revenue channels without dependence on fragmented third-party services.

1. Product Vision & Strategy

1.1 Vision Statement

Empower independent artists to build sustainable creative businesses by providing enterprise-grade tools for fan engagement and commerce in a single, integrated platform.

1.2 Business Goals

- Enable diversified revenue streams (events, music, merchandise)
- Increase direct-to-fan engagement and reduce platform intermediaries
- Provide actionable business intelligence for artist decision-making
- Scale from single artist usage to multi-artist marketplace

1.3 Success Metrics

- **Revenue Metrics:** Ticket conversion rate >12%, music revenue per fan >\$15, merchandise attach rate >25%
 - **Engagement Metrics:** Monthly active fans, repeat purchase rate >40%
 - **Platform Health:** Artist retention rate >85%, 99.9% uptime, transaction success rate >98%
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2. User Personas

2.1 Primary Users

Artist / Artist Manager

- Manages content, events, and product catalog
- Needs real-time analytics and revenue tracking
- Requires minimal technical expertise for platform operation

Fan / Customer

- Purchases tickets, streams music, buys merchandise
- Expects seamless cross-platform experience
- Values secure transactions and instant access to digital content

2.2 Secondary Users

Gate Staff

- Validates tickets at event entrances
- Requires offline-capable scanning with real-time sync
- Needs clear visual feedback for scan results

Platform Administrator

- Oversees system operations, compliance, and artist payouts
 - Monitors platform health and resolves disputes
 - Manages payment processing and settlements
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3. Product Scope

3.1 In Scope (MVP - Phase 1)

- Event creation, ticketing, and QR-based validation
- Music catalog management with download capability
- Basic merchandise store with inventory tracking
- Payment processing and artist settlements
- Core analytics dashboard

3.2 Phase 2

- Music streaming infrastructure
- Subscription-based access models
- Fan loyalty and rewards program
- Advanced bundling (ticket + merch + album)

3.3 Phase 3

- Multi-artist marketplace
- International expansion and multi-currency support
- Advanced analytics and predictive insights
- Seat mapping and venue visualization

3.4 Explicitly Out of Scope

- Third-party music distribution (Spotify, Apple Music integration)
 - NFT marketplace and blockchain features
 - Physical venue management systems
 - Artist collaboration tools
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4. Functional Requirements

4.1 Artist Profile & Dashboard

Profile Management

- Artist bio, profile images, and social media links
- Verification badge system with manual approval workflow
- Portfolio showcase (featured events, albums, products)

Unified Dashboard

- Real-time KPIs: ticket sales, music streams, merchandise orders
 - Revenue breakdown by category and time period
 - Upcoming events and pending tasks
 - Recent fan activity and engagement metrics
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4.2 Event Management

Event Creation

- Event metadata: title, description, category tags
- Venue selection or creation (name, address, capacity)
- Date/time configuration with timezone support
- Cover image and promotional materials upload
- Status management: Draft → Published → Sold Out → Completed → Cancelled

Ticket Configuration

- Multiple ticket types (General, VIP, VVIP, Early Bird)
- Flexible pricing with dynamic pricing rules (optional)
- Quantity limits per ticket type
- Sale window scheduling (start/end datetime)
- Refund policy definition per event

Ticket Delivery

- QR-code generation with encrypted payload
 - Email delivery with PDF attachment
 - In-app ticket wallet access
 - Transfer capability with security constraints
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4.3 Ticket Validation System

Scanner Application (Mobile/Web)

- QR code scanning with camera integration
- Offline validation mode with periodic sync
- Anti-fraud measures: single-use enforcement, duplicate detection
- Real-time attendance dashboard for event organizers

Validation Rules

- Prevent duplicate scans with immediate lock mechanism
- Validate ticket authenticity and event association
- Record scan metadata: timestamp, location, operator

4.4 Music Management

Catalog Structure

- Hierarchical organization: Albums → EPs → Singles → Tracks
- Metadata management: title, genre, ISRC, duration, lyrics
- Cover art upload with automatic resizing
- Release scheduling and embargo controls

Monetization Models

- Per-track or per-album purchases
- Free streaming with optional fan donations
- Subscription-based unlimited access (Phase 2)
- Bundle pricing strategies

Content Delivery

- Secure CDN-backed streaming URLs with expiring tokens
 - High-quality download options (MP3, FLAC)
 - DRM-free downloads with watermarking (optional)
 - Offline access in mobile applications
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4.5 Merchandising

Product Management

- Product types: physical (apparel, accessories) and digital (artwork, wallpapers)
- Variant system: size, color, material options
- SKU generation and inventory tracking
- Product images with multi-angle views

Inventory Management

- Real-time stock levels with low-stock alerts
- Inventory reservation during checkout
- Manual stock adjustments with audit logging

Order Fulfillment

- Delivery methods: standard shipping, express, event pickup
- Integration hooks for third-party fulfillment services
- Order status tracking: Pending → Processing → Shipped → Delivered
- Return and exchange workflow (Phase 2)

Bundling

- Cross-category bundles: Event + Merch, Event + Album, Album + Merch
 - Discounted bundle pricing with automatic calculation
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4.6 Fan Account & Commerce

User Account

- Single sign-on across all platform features
- Profile management and preferences
- Privacy settings and communication preferences

Digital Wallet

- Purchased tickets with QR codes
 - Music library with streaming and download access
 - Order history and tracking
 - Stored payment methods
 - Loyalty points balance (Phase 2)
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4.7 Payments & Financial Operations

Payment Processing

- Supported methods: Mobile Money (MTN, Airtel), Cards (Visa, Mastercard), Wallet balance
- PCI-DSS compliant payment gateway integration
- Multi-currency support with automatic conversion

Revenue Distribution

- Configurable platform commission (percentage or fixed fee)
- Automated artist payout scheduling (weekly, monthly)
- Tax withholding and reporting capabilities

- Transaction ledger with full audit trail

Refund Management

- Policy-based refund approvals
 - Partial and full refund support
 - Automatic reversal to original payment method
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4.8 Notifications & Communication

Transactional Notifications

- Order confirmations (email + SMS)
- Ticket delivery (email with PDF)
- Event reminders (24 hours, 2 hours before)
- Order shipping updates

Marketing Notifications

- New release announcements
- Event promotions
- Personalized recommendations
- Newsletter campaigns with segmentation

Notification Preferences

- User-controlled notification channels
 - Frequency management
 - Opt-out capabilities per category
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5. Non-Functional Requirements

5.1 Performance

- Support 10,000+ concurrent users during peak events
- Ticket scan response time <2 seconds (99th percentile)
- Page load time <3 seconds on 3G connections
- API response time <500ms (95th percentile)

5.2 Security

- JWT-based authentication with refresh tokens
- HTTPS enforcement across all endpoints
- Encrypted QR codes with time-limited validity
- Secure media delivery with signed URLs
- PCI-DSS Level 1 compliance for payment processing
- Regular security audits and penetration testing

5.3 Scalability

- Horizontal scaling for application servers
- Database read replicas for reporting queries
- CDN integration for media delivery
- Message queue for asynchronous processing
- Microservices architecture for core modules

5.4 Availability & Reliability

- 99.9% uptime SLA for ticketing and payment services
- Automated failover and disaster recovery
- Daily encrypted backups with 30-day retention
- Geographic redundancy for critical services

5.5 Compliance

- GDPR compliance for user data handling
 - Digital content rights verification workflow
 - Tax and VAT calculation with regional rules
 - Consumer protection and refund policy enforcement
 - Financial transaction audit logging
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6. Data Architecture

6.1 Design Principles

- Multi-tenant architecture with artist-level data isolation
- Strong referential integrity with foreign key constraints

- Event sourcing for financial transactions
- Optimized indexes for reporting and analytics
- Soft deletes for audit trail preservation

6.2 Core Entities

User Management

- users, artists, artist_verification

Event & Ticketing

- venues, events, ticket_types, tickets, scan_logs

Music Catalog

- albums, tracks, stream_logs, download_logs

Merchandising

- merch_products, merch_variants, inventory, merch_orders, merch_order_items

Commerce

- orders, payments, refunds, payouts

Engagement

- fan_wallets, notifications, user_preferences

6.3 Key Relationships

- Artist owns Events, Albums, and MerchProducts
- Event contains multiple TicketTypes
- TicketType generates individual Tickets
- Order aggregates Tickets and MerchOrderItems
- Payment links to Order with provider reconciliation

6.4 Performance Optimization

- Composite indexes: (event_id, status), (artist_id, created_at)
- Unique indexes: tickets.qr_token, payments.provider_reference
- Partitioning: stream_logs by month, scan_logs by event_id
- Caching: ticket validation state, artist catalogs, venue data

7. Analytics & Reporting

7.1 Artist Analytics

- **Sales Performance:** Revenue by category, conversion funnels, discount impact
- **Event Analytics:** Ticket sales velocity, attendance rates, gate scan analytics
- **Music Insights:** Streams by track, geographic distribution, listening patterns
- **Merchandise Reports:** Best-selling products, inventory turnover, variant performance
- **Fan Demographics:** Age, location, purchase behavior, lifetime value

7.2 Platform Analytics

- Transaction volumes and gross merchandise value (GMV)
 - Active artists and fan growth metrics
 - Payment success rates and method distribution
 - System performance and error rates
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8. Technical Integration Points

8.1 Payment Gateways

- Mobile Money: MTN MoMo API, Airtel Money API
- Card Processing: Stripe, Flutterwave
- Wallet System: Internal ledger with transaction logging

8.2 Communication Services

- Email: SendGrid, AWS SES
- SMS: Twilio, Africa's Talking
- Push Notifications: Firebase Cloud Messaging

8.3 Media Services

- CDN: Cloudflare, AWS CloudFront
 - Storage: AWS S3, Azure Blob Storage
 - Transcoding: AWS MediaConvert (for future video)
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9. Risk Management

Risk	Impact	Likelihood	Mitigation Strategy
Ticket fraud and duplication	High	Medium	Encrypted QR codes, scan locking, real-time validation
Music piracy and unauthorized sharing	High	High	Secure streaming URLs, watermarking, DRM options
Payment gateway failures	Critical	Low	Multiple provider fallback, retry logic, status monitoring
Logistics and fulfillment delays	Medium	Medium	Partner SLA management, tracking integration, communication
Data breach or security incident	Critical	Low	Regular audits, encryption, access controls, incident response plan

10. Release Roadmap

MVP (Q1 2026)

- Artist profiles and dashboard
- Event creation and ticketing
- QR-based validation
- Music downloads
- Basic merchandise store
- Payment processing
- Core analytics

Phase 2 (Q2-Q3 2026)

- Music streaming infrastructure
- Subscription models
- Loyalty and rewards program
- Advanced bundling
- Mobile applications (iOS, Android)

Phase 3 (Q4 2026+)

- Multi-artist marketplace
 - International expansion
 - Seat mapping and venue visualization
 - Live streaming integration
 - Predictive analytics and ML recommendations
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11. Acceptance Criteria

MVP Launch Criteria

- 10 beta artists successfully onboarded
 - 100+ tickets sold and validated without fraud incidents
 - 500+ music downloads processed
 - Payment success rate >95%
 - Zero critical security vulnerabilities
 - Artist satisfaction score >8/10
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12. Dependencies & Assumptions

Dependencies

- Payment gateway partnerships and API access
- Mobile Money aggregator agreements
- CDN and cloud infrastructure provisioning
- Legal compliance review and approval

Assumptions

- Artists have existing fan bases for initial traction
 - Reliable internet connectivity for primary use cases
 - Payment gateway uptime >99%
 - Artist willingness to adopt new platform
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13. Open Questions

1. What is the platform commission structure (percentage vs fixed fee)?
 2. Should the platform handle physical merchandise fulfillment or partner with third parties?
 3. What are the tier structures for artist subscription plans (if applicable)?
 4. How should we handle international tax compliance for digital goods?
 5. What level of customization should artists have over their storefronts?
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Appendix A: Database Schema Summary

Critical Tables

- **users:** Authentication and role management
- **artists:** Artist profiles and verification status
- **events:** Event metadata and lifecycle
- **tickets:** Individual ticket instances with QR tokens
- **orders:** Purchase aggregation and payment linking
- **payments:** Payment provider reconciliation
- **stream_logs:** Music consumption tracking
- **inventory:** Real-time merchandise stock levels

Key Constraints

- One-to-one: users ↔ artists
 - One-to-many: events → ticket_types → tickets
 - Many-to-many: orders ↔ tickets (through order_items)
 - Soft deletes: orders, payments, tickets (for audit trail)
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Document Control

Version	Date	Author	Changes
1.0	Jan 2026	Kubwimana David	Initial draft
1.1	Jan 2026	Claude	Structural improvements, clarity enhancements

Approval Signatures

Role	Name	Signature	Date
Product Owner			
Technical Lead			
Stakeholder			